

Overleaf (LaTeX) – Tags with Equation Writing Examples

Inline Math

$a^2 + b^2 = c^2$ → written as: `$a^2 + b^2 = c^2$`

Display Math

$E = mc^2$ → written using `\[ ... \]`

Fraction

$\frac{a}{b}$ → `\frac{a}{b}`

Square Root

$\sqrt{x}$ → `\sqrt{x}`

Summation

$\sum_{i=1}^n i$ → `\sum_{i=1}^n i`

Integral

$\int_0^1 x^2 dx$ → `\int_0^1 x^2 dx`

Limit

$\lim_{x \rightarrow 0} \frac{\sin x}{x}$ → `\lim_{x \rightarrow 0} \frac{\sin x}{x}`

Sine

$\sin x$ → `\sin x`

Cosine

$\cos x$ → `\cos x`

Tangent

$\tan x$ → `\tan x`

Logarithm

$\log x$ → `\log x`

Natural Log

$\ln x$ → `\ln x`

Exponential

e^x → `e^x`

Aligned Equations

$$\begin{aligned} x+y&=10 \\ x-y&=4 \end{aligned}$$

Matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Set Notation

$x \in A \subset B$ → `\[ x \in A \subset B \]`

Infinity

$x \rightarrow \infty$ → `\[ x \rightarrow \infty \]`

Greek Letters

$\alpha + \beta = \theta$ → `\[ \alpha + \beta = \theta \]`